

Validator-Guided Repair on a Shared Initial LLM Candidate: A Preregistered Paired Evaluation for Network Configuration Safety

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CNSM 2026 Agentic AI Researcher Track

Abstract—We preregistered and executed a paired confirmatory experiment (PREREG-CAND-001-VER-GVR-2026-09-01) to measure whether permitting a deterministic network-configuration validator plus one automated repair changes the rate of validator-valid executable configurations relative to leaving the identical sampled LLM candidate unguarded (`shared_initial_candidate` semantics). Using the declared generator `gpt-5-mini` (hosted adapter `openai_responses`) and deterministic `netops_validator_v5` on $N = 200$ paired intents (`completed_episode_count = 400`; `model_calls_used = 200`), every shared initial candidate passed validator checks and no repairs were attempted or applied (`repair_attempt_count = 0`; `repair_applied_count = 0`). Observed outcomes: `baseline_success = 200/200`, `guarded_success = 200/200`; paired contingency $n11 = 200$, $n10 = 0$, $n01 = 0$, $n00 = 0$; exact McNemar two-sided $p = 1.0$; paired bootstrap percentile 95% CI (seed = 1729; 10,000 resamples) = $[0.0, 0.0]$. We present artifact-grounded methodology, execution accounting, representative validator diagnostics, compact contingency and repair summaries, and a focused discussion clarifying the causal limits of the executed estimand under `shared_initial_candidate` semantics and preregistered follow-on designs required to measure repair efficacy under independent sampling, higher difficulty, or adversarial inputs.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration sequences in network operations (NetOps). Erroneous sequences, syntactic mistakes, or fabricated fields can cause service disruption or policy violations when applied to devices. A common architectural mitigation is a generate→validate→repair (GVR) pipeline: generate candidate configuration(s), deterministically validate them against intent/topology/workflow constraints, and trigger automated repair when the validator finds faults. Prior work documents verifier-guided improvements in infrastructure-as-code and related code-generation domains and recommends grounding strategies to reduce hallucination in operational tasks [1–3]. However, whether verifier-guided pipelines reduce execution-level operational risk for network configurations remains an open empirical question because prior demonstrations often measure textual correctness or IaC test suites rather than executed device-state safety in packet-level or emulator contexts [3,4].

This paper reports a fully preregistered paired confirmatory experiment that isolates the downstream causal contribution of a deterministic validator and any permitted repair acting on the

exact same sampled LLM candidate (`shared_initial_candidate` semantics). Under these semantics baseline and guarded episodes for each intent begin from the identical initial generation; any paired difference can therefore only arise from deterministic validation and a permitted repair of that same candidate. Our primary research question is: under `shared_initial_candidate` semantics, does permitting a deterministic validator plus at most one automated repair change the proportion of validator-valid executable configurations relative to leaving the same initial candidate unguarded? The contribution is an artifact-grounded report of the executed estimand with complete execution accounting, exact inferential statistics, representative validator diagnostics, and precise recommendations for preregistered follow-on designs that can measure repair efficacy when independent sampling, elevated difficulty, or adversarial inputs are employed.

II. RELATED WORK

Hallucination and factual unreliability in LLM outputs are well documented and motivate grounding, retrieval, and verification strategies for operational tasks [1]. Several recent surveys synthesize these failure modes and recommend evidence-grounding or verification as primary mitigations; they emphasize that hallucination manifests across domains and that mitigation effectiveness depends strongly on task framing and the grounding mechanism used [1].

Retrieval-augmented and evidence-grounded pipelines have been shown to reduce hallucination in operations-style question answering and documentation tasks, primarily by constraining the model with external factual context and by post-generation grounding checks [2]. These evaluations typically measure textual fidelity (e.g., citation correctness, answer grounding) rather than executable device-state safety. As a result, reductions in text-level hallucination do not directly imply reductions in execution risk when generated artifacts are translated into device CLI and applied to network state.

Generate→validate→repair (GVR) architectures have been proposed and demonstrated in adjacent domains such as infrastructure-as-code (IaC) and program synthesis, where deterministic or policy-guided verifiers detect rule violations and either supply feedback to the generator or guide repair loops [3]. TerraFormer and related verifier-guided IaC systems operationalize a feedback loop that fine-tunes or constrains generation via verifier signals; reported gains are measured against

IaC test suites and policy checks rather than execution on heterogeneous device families. Key methodological elements highlighted by these works include (a) verifier expressivity (what constraints and dependency rules are implemented), (b) repair strategy (single corrective edit, multi-step rewriting, or fine-tuning), and (c) how verifier feedback is integrated (in-prompt, tool call, or offline retraining) [3].

NetOps-specific evaluations of LLM capability concentrate on intent translation accuracy, syntax compliance, and component-level parsing rather than end-to-end execution safety in simulators or testbeds [4]. These studies typically report metrics such as command-level BLEU/similarities, slot accuracy, or human adjudication of intent fidelity; they provide useful capability baselines but leave open how textual errors map to executable misconfigurations or service disruption when applied to device state. AIOps surveys further underscore the gap between capability benchmarking and execution-aware evaluation: they call for standardized execution-level benchmarks and validator fidelity measurements that tie generator outputs to concrete operational safety outcomes [5].

From a methodological standpoint, prior literature therefore separates three evaluation axes: (1) generator sampling variability (how often first-pass candidates are correct), (2) validator fidelity (whether the verifier captures the operational invariants and dependency rules relevant to execution), and (3) repair efficacy (whether permitted repairs transform invalid candidates into valid, executable ones). Many published demonstrations conflate (1) and (3) by sampling independently across conditions or by evaluating repaired outputs only at the text level. The paired, shared_initial_candidate semantics used in our preregistration explicitly isolate axis (2) and (3) acting on a single sampled candidate: baseline and guarded arms begin from the identical generation so any difference must come from deterministic validation and permitted repair rather than from independent sampling.

This distinction clarifies why IaC results are informative but not decisive for NetOps execution safety. TerraFormer-style verifier-guided improvements (IaC) indicate that verifier feedback can materially improve generator correctness in structured code domains [3]. However, network device configuration introduces additional complexity: vendor CLI differences, runtime protocol interactions, timing-sensitive sequences, and multi-device workflows. Existing NetOps capability studies and AIOps surveys document translation ability and recommend validator uplift but, crucially, do not provide standardized execution-level outcomes that map generator errors to runnable device states—creating the empirical gap that motivated our execution-focused preregistered study [4,5].

Finally, the literature highlights practical evaluation considerations that our experiment operationalizes: use deterministic task manifests and contamination checks to avoid corpus leakage, record provider audit traces and model configuration details (including explicitly noting an unset temperature rather than assuming deterministic sampling), and archive per-episode validator traces (validation_before/validation_after, parsed/required command counts,

repair_applied flags) to permit exact replay and independent verification [2–5]. These provenance and scoring practices are commonly recommended but rarely presented end-to-end in a preregistered execution bundle that also enforces a narrowly scoped estimand—our study adopts these artifact practices to make the estimand and its limits explicit and reproducible.

III. METHODOLOGY

Preregistration and primary estimand. The study preregistration (PREREG-CAND-001-VER-GVR-2026-09-01) froze the primary estimand and analysis before any model calls. Primary estimand: `paired_success_rate_difference_guarded_minus_baseline` (per-intent paired difference in proportion of validator-valid executable configurations) under `shared_initial_candidate` semantics. Design: deterministic paired sample of `task_count = 200` intents producing `planned_episode_count = 400` episodes. Generation semantics: exactly one sampled initial generation per pair (shared) and deterministic reuse by baseline and guarded episodes. Guarded episodes allowed up to one automated repair (`maximum_repair_calls_per_task = 1`). Primary test: exact two-sided McNemar test; paired bootstrap (seed = 1729; 10,000 resamples) for a percentile 95% CI.

Task bank and contamination controls. A deterministic task generator produced the manifest of 200 synthetic `intent_configuration_repair_v1` tasks. Each manifest entry records `initial_state`, `required_sequence`, difficulty annotations, and a `generation_seed` for deterministic replay. Preexecution contamination checks (SHA-256 exact-match and n-gram similarity thresholds) were applied; no exact-match contamination exclusions were flagged and the confirmatory sample executed intact. Contamination and task manifest artifacts are archived in the execution evidence bundle.

Model calls, generation semantics, and provider audit. Execution used the declared generator labeled `gpt-5-mini` via the hosted adapter `openai_responses` and consumed `model_calls_used = 200` (one shared initial generation per pair). The archived `model_configuration.json` records `declared_model_version = "gpt-5-mini"` and `temperature = null/unset`; we explicitly state that `null/unset` is not evidence of deterministic sampling and does not imply `temperature = 0`. Provider audit traces record `hosted_model_call_event_count = 200`, `cache_miss_count = 200`, and `stage_counts` showing `shared_initial_generation = 200` consistent with the preregistered `shared_initial_candidate` semantics.

Deterministic validator and repair policy. Scoring used `deterministic_netops_validator_v5` with `scoring_rule = "full_intent_constraint_validation."` The validator parses candidate commands, applies task `workflow_policies` and `required_sequence` checks, and produces structured validator traces with `validation_before` and `validation_after` subtables, `parsed_command_count`, `required_command_count`, `normalized_configuration`, `violation_codes`, and a `repair_applied` flag. For guarded episodes one automated repair attempt was permitted; the repair, if invoked, would produce

a new candidate which the validator would re-validate. Under `shared_initial_candidate` semantics only the guarded arm could alter the identical initial candidate via the permitted repair; the baseline arm reused the same candidate unchanged.

Mapping to outcomes and missingness. `Validator.valid = true` mapped to a binary "executable/valid" outcome for each episode. Primary analysis adhered to the preregistered `complete_pair_primary` policy (exclude incomplete pairs). A sensitivity analysis (failure-as-zero) was preregistered to treat missing or failed episodes as unsafe. Multiplicity: single confirmatory McNemar test; exploratory subgroup analyses (e.g., by difficulty patterns) were labeled exploratory and corrective procedures (Benjamini–Hochberg) were specified if applied.

IV. RESULTS

Execution accounting and deterministic reconciliation. Execution completed per the manifest: `planned_episode_count = 400`, `completed_episode_count = 400`, `failed_episode_count = 0`, `model_calls_used = 200` (execution/manifests archived). Analysis missingness_summary reports `complete_pairs = 200` and zero failed or unscorable episodes. Deterministic reconciliation shows `stage_counts = {shared_initial_generation: 200}` and `cross_condition_cache_key_reuse_observed = false`; all deterministic consistency checks passed.

Primary outcomes and contingency. Condition summaries: `baseline_successes = 200/200` and `guarded_successes = 200/200` (`success_rate = 1.0` each). Paired contingency: `n11 = 200`, `n10 = 0`, `n01 = 0`, `n00 = 0`. Exact McNemar two-sided $p = 1.0$. Paired bootstrap (seed = 1729; 10,000 resamples) percentile 95% CI for the paired difference = [0.0, 0.0]. These artifacts are archived as `analysis/condition_summary.csv` and `analysis/paired_contingency_table.csv`.

Repair activity and validator diagnostics. `Repair_attempt_count = 0` and `repair_applied_count = 0` across the confirmatory sample: the validator flagged no violations on any shared initial candidate and therefore the permitted guarded repair path was never invoked. Representative per-episode JSON artifacts (examples archived under `execution/scoring/`, e.g., `task-000004-baseline.json` and `task-000004-guarded.json`) show identical `shared_initial_candidate` content and identical `validation_after.normalized_configuration` values. Example shared initial candidate (task-000004):

```
interface edge2 admin down
interface edge2 vlan 40 interface edge2 admin up
interface edge1 admin down
```

For that example `parsed_command_count = 4`, `required_command_count = 4`, and `validator.valid = true`. Similar diagnostics hold for sampled representative tasks (task-000005, task-000006).

Contamination and provider audit summary. Pre-execution contamination checks flagged no exact matches; `analysis/contamination_summary.csv` is empty. Provider audit recorded `hosted_model_call_event_count = 200` and `cache_miss_count = 200`. No infrastructure failures or retrials occurred; `analysis/analysis_log.jsonl` documents the bootstrap

settings used (`bootstrap_resamples = 10000`; `bootstrap_seed = 1729`) and that paired analysis completed successfully.

Compact repair summary (explicit). Table 2 (below) provides the preregistered reviewer request summary derived from archived artifacts: `total_pairs = 200`; `repair_attempt_count = 0`; `repair_applied_count = 0`. The absence of any validator detections fully explains the observed degenerate contingency: with no validator detections, the guarded repair path had no opportunity to alter outcomes.

V. DISCUSSION

Interpreting the executed estimand. The `shared_initial_candidate` semantics were intentionally chosen to isolate the causal effect of deterministic validation and any permitted repair acting on the exact same generator output. The executed estimand therefore answers a narrowly defined causal question: when baseline and guarded episodes begin from the identical sampled candidate, can deterministic validation and at most one automated repair change the validator-valid/executable outcome? The observed null effect (paired difference = 0.0) shows that, for this task bank and this single sampled candidate per pair, the validator found no violations and the permitted repair path was never exercised. Importantly, this does not evaluate whether guarded pipelines reduce first-pass generator error rates under independent sampling or different model temperatures; such claims would require a different preregistered estimand and execution.

Threats to validity (artifact-grounded). Internal validity: the paired `shared_initial_candidate` design eliminates sampling variability across arms and thus preserves attribution of any observed repair effect to validator/repair alone; however, this design precludes measurement of generator-level effects. Construct validity: `deterministic_netops_validator_v5` enforces syntactic and workflow-policy checks and maps results to a boolean valid flag; it is an abstraction that does not execute vendor CLI on physical hardware nor perform packet-level emulation—dynamic protocol/timing effects are therefore not represented. External validity: results are conditioned on one declared generator (gpt-5-mini via adapter `openai_responses`), a synthetic deterministic task bank, and one validator implementation; generalization beyond these artifacts is unsupported. Conclusion validity: zero discordant outcomes preclude inference about repair benefit magnitude under different sampling regimes because the guarded path was not invoked.

Operational implications and preregistered follow-on designs. The null result here should not be read as evidence that GVR pipelines are unnecessary in practice. Instead it demonstrates an estimand boundary: under `shared_initial_candidate` semantics and the executed task bank the guarded pipeline had no opportunity to change outcomes. To empirically evaluate repair efficacy in operationally relevant regimes, we preregistered and recommend the following estimand/design modifications (each reuses archived artifacts where applicable): (A) remove `shared_initial_candidate` semantics so baseline and guarded arms receive independent first-pass samples (this permits the repair path to act on generator faults but requires

a revised paired/unpaired analysis plan to separate generation variability from repair effects); (B) select or synthesize tasks labeled "extreme" difficulty or inject adversarial perturbations to raise expected validator failure rates and thereby exercise repair logic; (C) preregister non-null temperature values and temperature sweeps to evaluate sample diversity versus validator-failure tradeoffs; (D) increase validator fidelity by integrating vendor CLI semantics or packet-level emulation so repairs are evaluated against dynamic runtime behavior; (E) expand to multi-model sampling to assess generality across generator families.

Reproducibility and artifact utility. All per-episode scoring JSON artifacts include validator traces with `validation_before/validation_after` subtables, `normalized_configuration`, `parsed_command_count`, `required_command_count`, and `repair_applied` flags permitting independent verification that repairs did not run. Execution and analysis artifacts (`execution/raw_results.jsonl`, `execution/task_manifest.jsonl`, `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl`) are archived in the evidence bundle and enable replay of the exact `shared_initial_candidate` sampling. The `model_configuration.json` records `declared_model_version = "gpt-5-mini"` and `temperature = null`; `null/unset` must not be interpreted as deterministic sampling. The master prompt immutable reference is archived (see Disclosure). These artifacts are sufficient to reproduce the executed estimand and deterministic reconciliation.

VI. LIMITATIONS

- `Shared_initial_candidate` semantics: the design produced exactly one sampled initial generation per intent and reused it across baseline and guarded conditions; this measures validator/repair effects only and cannot detect differences caused by independent per-condition sampling or prompt engineering.
- Temperature `unset` (`temperature = null`) in the archived model configuration: `null/unset` is not evidence of deterministic sampling and does not imply `temperature = 0`; sampling nondeterminism may still have occurred during the single sampled generation.
- Single model and single hosted provider: results reflect `gpt-5-mini` under the archived adapter; generalization to other models or model families is unsupported by these artifacts.
- Synthetic task bank and validator abstraction: tasks were synthetic `'intent_configuration_repair_v1'` items and the deterministic validator is an abstraction; realistic vendor CLI idiosyncrasies, emergent runtime interactions, and hardware effects were not executed on live devices.
- No observed validator failures: all initial candidates were validator-valid, so the experiment could not assess the repair step's effectiveness; the null effect therefore reflects the task bank and sampling semantics rather than evidence that GVR is unnecessary in general.

VII. CONCLUSION

We executed a fully preregistered paired confirmatory experiment that measures the downstream effect of a deterministic validator plus an allowed single automated repair on the exact same sampled LLM candidate per intent (`shared_initial_candidate` semantics). Using the declared generator `gpt-5-mini` and deterministic `netops_validator_v5` on 200 paired intents, every shared initial candidate passed validation and no repairs were attempted or applied; guarded and baseline outcomes were identical for the executed estimand (paired difference = 0.0; McNemar $p = 1.0$; paired bootstrap CI = [0.0, 0.0]). This artifact-grounded null result demonstrates an estimand boundary: under `shared_initial_candidate` semantics and the executed task bank the guarded pipeline could not be exercised and therefore could not change outcomes. It does not imply that GVR pipelines are unnecessary generally. Preregistered follow-on experiments that (1) remove `shared_initial_candidate` semantics, (2) increase task difficulty or inject adversarial inputs, (3) set explicit sampling parameters, and (4) raise validator fidelity are required to empirically evaluate repair efficacy under realistic sampling variability and adversarial conditions. The archived artifacts and reconciliation files enable exact reproduction of the executed estimand and provide a secure base for precise preregistered extensions.

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One-line reiteration of the primary estimand limit: the preregistered primary estimand intentionally used `shared_initial_candidate` semantics; evaluating per-condition independent sampling requires a different preregistration and analysis plan (explicitly recommended above).

DISCLOSURE STATEMENT

Disclosure (concise): The human Research Director selected the broad topic family (Generative AI / LLMs for NetOps) before the autonomy lock. After the autonomy lock the autonomous system performed literature review, experiment selection, preregistration (PREREG-CAND-001-VER-GVR-2026-09-01), code generation, execution, deterministic validation, automated analysis, and manuscript drafting without manual human editing of technical content. Declared model used in execution: `gpt-5-mini` via hosted adapter `'openai_responses'` (execution used `model_calls_used = 200`; archived `model_configuration.json` records `temperature = null/unset`;

null/unset is not evidence of deterministic sampling and does not imply temperature = 0). Generation semantics executed: `shared_initial_candidate` — exactly one sampled initial generation per intent was produced and deterministically reused for both baseline and guarded conditions. Validator: `deterministic_netops_validator_v5` scored candidates using `'full_intent_constraint_validation'` and a single automated repair iteration was permitted in guarded episodes; in this run the validator flagged no violations and no repairs were applied (`repair_attempt_count` = 0; `repair_applied_count` = 0). Execution accounting: `completed_episode_count` = 400 (200 paired intents) completed without preregistration deviations. Master prompt immutable reference: `provenance/master_prompt.txt` — SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba. Pre-lock infrastructure development and rehearsal occurred but pre-lock artifacts were not used as empirical evidence in this final run. After the autonomy lock no human manually rewrote technical results, manuscript text, figures, or references.

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